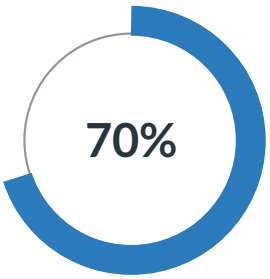
Results

Carter Feng — 1st Attempt



70
Out of 100 points

29:27:23
Time for this attempt

This assessment is locked.

Attempt History

Results	Points	Score	(First score is kept)
Attempt 1	70 of 100	70%	(First score)

Your Answers:

1 — 10 / 10 points — Matching

```
Create a Red Black Binary Search Tree in the following order:
221 11 779 73 327 215 422 799 469 159 then delete 11 327 422 469 th

What are the following values at the 2nd rotation when creating the
```

Grandparent Node	✓ 779

Parent Node	✓	327
X Node	✓	422
Type of Rotation	✓	left right

2 0 / 10 points Matching

Create a Red Black Binary Search Tree in the following order:
221 11 779 73 327 215 422 799 469 159 then delete 11 327 422 469 th

What are the following values at the 4th color swap when creating t

Parent Node	✗	799
		Correct Answer: **779**
Parent's Left Node	✗	536
		Correct Answer: **765**
Parent's Right Node	✗	(no answer)
		Correct Answer: **799**

3 10 / 10 points Fill in the Blank

Create a Red Black Binary Search Tree in the following order:
221 11 779 73 327 215 422 799 469 159 then delete 11 327 422 469 th

What is the number of degree 2 nodes in the red black tree after co

✓ 4

4 10 / 10 points Fill in the Blank

Create a Red Black Binary Search Tree in the following order:
221 11 779 73 327 215 422 799 469 159 then delete 11 327 422 469 th

What is the root of the red black tree after completion of the tree

✓ 221

5 10 / 10 points Matching

Create a Red Black Binary Search Tree in the following order:
221 11 779 73 327 215 422 799 469 159 then delete 11 327 422 469 th

What are the following values at the 1st color swap when creating t

Parent Node	✓ 221
Parent's Left Node	✓ 11
Parent's Right Node	✓ 779

6 10 / 10 points Matching

Create a Red Black Binary Search Tree in the following order:
221 11 779 73 327 215 422 799 469 159 then delete 11 327 422 469 th

What is the first ten values in a preorder traversal after 765 is d

1	2	3	4	5	6	7	8	9	10

1	✓ 221
2	✓ 126
3	✓ 73
4	✓ 106
5	✓ 159
6	✓ 130
7	✓ 621
8	✓ 312
9	✓ 306
10	✓ 496

7 0 / 10 points Matching

Create a Red Black Binary Search Tree in the following order:
221 11 779 73 327 215 422 799 469 159 then delete 11 327 422 469 th

What are the following values at the 6th rotation when creating the

Grandparent Node	✗ 779	Correct Answer: **306**
Parent Node	✗ 469	Correct Answer: **536**
X Node	✗ 621	Correct Answer: **496**
Type of Rotation	✗ left left	Correct Answer: **right left**

8 0 / 10 points Matching

Create a Red Black Binary Search Tree in the following order:
221 11 779 73 327 215 422 799 469 159 then delete 11 327 422 469 th

What are the following values at the 1st special case deletion when

Parent Node	✗ (no answer)	Correct Answer: **496**
Siblings's Left Child	✗ (no answer)	Correct Answer: **306**
Siblings's Right Child	✗ (no answer)	Correct Answer: **420**
Sibling	✗ (no answer)	Correct Answer: **312**

9 10 / 10 points Matching

```
Create a Red Black Binary Search Tree in the following order:
221 11 779 73 327 215 422 799 469 159 then delete 11 327 422 469 th

What is the last ten values in a preorder traversal after 765 is de
```

1 2 3 4 5 6 7 8 9 10

8	✓	420
9	✓	779
10	✓	799

1	✓	106	
2	✓	159	
3	✓	130	
4	✓	621	
5	✓	312	
6	✓	306	
7	✓	496	

10 10 / 10 points Matching

Create a Red Black Binary Search Tree in the following order:
221 11 779 73 327 215 422 799 469 159 then delete 11 327 422 469 th

What are the values on level 1 of the completed red black tree?

1 2

1	✓	126	
2	✓	621	